

Final Model and Evaluation Setup

Setup

- ▶ Collector-disjoint train/validation/test split
- ▶ Majority-vote segment labels from $[T, C, A]$ annotations
- ▶ Macro average precision used for model selection
- ▶ Final model: validation-selected MLP probability ensemble

Why macro AP

- ▶ Threshold-free for multi-label sigmoid outputs
- ▶ Gives rare classes equal weight
- ▶ Avoids hiding light_switch/bell errors behind common classes

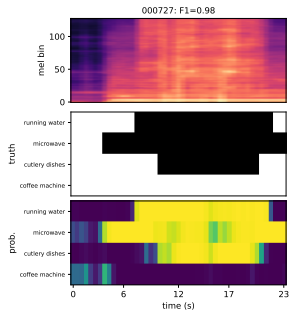
Model	Macro AP	Micro AP
Prior	0.064	0.121
Logistic	0.529	0.664
MLP ensemble	0.638	0.754

Held-out test set; macro / micro average precision.

Qualitative Case Studies

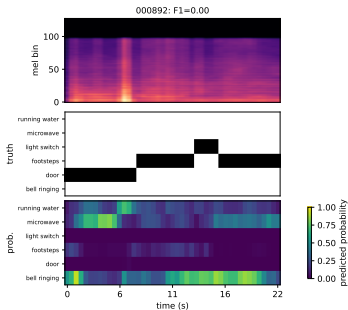
Success: file 000727

- ▶ Kitchen (microwave, water, dishes); static phone
- ▶ File-level F1: **0.978**
- ▶ Sustained events track mel-spectrogram energy



Failure: file 000892

- ▶ Hallway (door, footsteps, light switch); mobile phone
- ▶ File-level F1: **0.000**
- ▶ Short transients shifted or confused



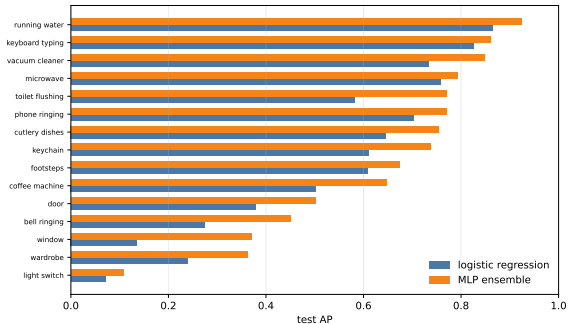
What the Classifier Detects Reliably

Reliable classes:

- ▶ running_water: 0.925 AP
- ▶ keyboard_typing: 0.861 AP
- ▶ vacuum_cleaner: 0.847 AP
- ▶ microwave: 0.793 AP

Interpretation:

- ▶ Sustained, repeated evidence across adjacent segments
- ▶ Distinct broadband or periodic spectral signatures
- ▶ Temporal context window helps stabilize predictions



Failure Modes and Next Steps

1. **Short transients:** light_switch remains hardest (0.108 AP); one-second aggregation and half-second hops can shift the evidence away from the label segment.
2. **Mechanical confusions:** door, window, and wardrobe_drawer share impact, hinge, and handling sounds; the feature representation separates them only weakly.
3. **Rare-class instability:** Few positive examples make AP sensitive to label noise and missed annotator events.

Most promising improvements: raw-audio temporal models, focal loss or rare-class sampling, explicit event-duration priors, and post-processing that smooths isolated false alarms without erasing real short events.